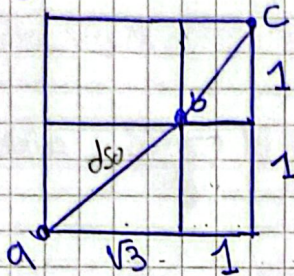


Parcial 2 2024

Ej 1) Determina el cambio de longitud que sufre la fibra $a\bar{b}c$ que está sujeta por $a\bar{b}$ y $\bar{b}c$ sometida a pequeñas deformaciones $\epsilon = 10^{-3} \begin{bmatrix} 10 & 5 \\ 5 & 20 \end{bmatrix}$



Diferencia entre longitudes entre dos puntos vecinos

$$ds^2 - dso^2 = 2 \epsilon_{ij} da_i da_j$$

El problema aclara que en pequeñas deformaciones, el alargamiento ΔL es infinitesimal

Longitud Final es la inicial + el alargamiento $ds = dso + \Delta L$

$$ds^2 - dso^2 = 2 \epsilon_{ij} da_i da_j \quad \text{Factorizamos}$$

$$(ds - dso)(ds + dso) = 2 \epsilon_{ij} da_i da_j$$

Lo que buscamos ΔL $\approx 2dso$

$$\Delta L \cdot 2dso = 2 \epsilon_{ij} da_i da_j$$

$$[\Delta L dso = \epsilon_{ij} da_i da_j] \quad (1)$$

Tramo $a\bar{b}$: • Para ir de $a \rightarrow b$ nos movemos $\sqrt{3}$ en x } $da = (\sqrt{3}; 1)$
 1 en y } $da_1 \quad da_2$

$$\bullet dso = \text{hipotenusa } H^2 = C^2 + C^2 \rightarrow dso = \sqrt{(\sqrt{3})^2 + 1^2} = 2$$

$$\bullet \epsilon_{ij} da_i da_j = 10^{-3} [\epsilon_{11} da_1 da_1 + \epsilon_{12} da_1 da_2 + \epsilon_{21} da_1 da_2 + \epsilon_{22} da_2 da_2]$$

las iguales = $2\epsilon_{12} da_1 da_2$

$$\epsilon_{ij} da_i da_j = 10^{-3} [10(\sqrt{3})^2 + 2 \cdot 5 \cdot \sqrt{3} \cdot 1 + 20 \cdot 1 \cdot 1]$$

$$\epsilon_{ij} da_i da_j = 10^{-3} [50 + 10\sqrt{3}] \quad \text{reemplazamos en (1)}$$

$$\Delta L_{ab} \cdot 2 = 10^{-3} [50 + 10\sqrt{3}]$$

$$[\Delta L_{ab} = 10^{-3} [25 + 5\sqrt{3}]]$$

$$\text{Trans } \vec{bc} = da_1 = 1 \quad da_2 = 1 \quad ds_0 = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\epsilon_{ij} da_i da_j = \epsilon_{11} da_1 da_1 + 2\epsilon_{12} da_1 da_2 + \epsilon_{22} da_2 da_2$$

$$\epsilon_{ij} da_i da_j = 10^{-3} [10 \cdot 1^2 + 2 \cdot 5 \cdot 1 \cdot 1 + 20 \cdot 1^2]$$

$$\epsilon_{ij} da_i da_j = 40 \cdot 10^{-3} \quad \text{Recombination (1)}$$

$$\Delta L_{bc} \cdot \sqrt{2} = 40 \cdot 10^{-3}$$

$$\Delta L_{bc} = 20\sqrt{2} \times 10^{-3}$$

$$\Delta L = \Delta L_{ab} + \Delta L_{bc}$$

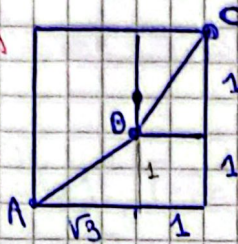
$$\Delta L = (10^{-3} \cdot [25 + 5\sqrt{3}]) + 10^{-3} \cdot 20\sqrt{2}$$

$$\Delta L = 0,061944$$

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Pequeñas deformaciones

Ej 11



Determinar el cambio de longitud de $\overline{abc}$ que está compuesta por los rectos $\overline{ab}$ y $\overline{bc}$

$$\epsilon = 10^{-3} \begin{bmatrix} 10 & 5 \\ 5 & 20 \end{bmatrix}$$

Para Pequeñas deformaciones el cambio de longitud de un segmento es: $\Delta L = \frac{V^T \epsilon V}{L}$

Tramo 1 recto a-b. $a = (0,0)$ $b = (\sqrt{3}, 1)$ $V_{ab} = b - a = (\sqrt{3}, 1)$
 rectos normalizados

$$\text{longitud} = H^2 = C^2 + C^2 \rightarrow L_{AB} = \sqrt{(\sqrt{3})^2 + 1^2} = 2$$

$$\Delta L_{ab} = \frac{V_{ab}^T \cdot \epsilon \cdot V_{ab}}{L_{AB}}$$

$$\epsilon \cdot V_{ab} = 10^{-3} \begin{bmatrix} 10 & 5 \\ 5 & 20 \end{bmatrix} \cdot \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} = 10^{-3} \begin{bmatrix} 5 + 10\sqrt{3} \\ 20 + 5\sqrt{3} \end{bmatrix}$$

$$\Delta L_{ab} = \frac{[\sqrt{3}, 1] \cdot 10^{-3} \begin{bmatrix} 5 + 10\sqrt{3} \\ 20 + 5\sqrt{3} \end{bmatrix}}{2} = \frac{10^{-3} [\sqrt{3} \cdot (5 + 10\sqrt{3}) + 1 \cdot (20 + 5\sqrt{3})]}{2}$$

$$[\Delta L_{ab} = 0,03366]$$

Tramo 2 b-c. $b = (\sqrt{3}, 1)$ $c = (\sqrt{3} + 1, 2)$ $V_{bc} = c - b = (1, 1)$

$$L_{bc} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\Delta L_c = \frac{V_{bc}^T \cdot \epsilon \cdot V_{bc}}{L_c}$$

$$\epsilon \cdot V_{bc} = 10^{-3} \begin{bmatrix} 10 & 5 \\ 5 & 20 \end{bmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{bmatrix} 15 \\ 25 \end{bmatrix} \cdot 10^{-3}$$

$$\Delta L_c = \frac{(1 \ 1) \cdot 10^{-3} \begin{bmatrix} 15 \\ 25 \end{bmatrix}}{\sqrt{2}} = \frac{10^{-3} (15 + 25)}{\sqrt{2}}$$

$$[\Delta L_c = 0,02828]$$

$$[\Delta L = \Delta L_{ab} + \Delta L_{bc} = 0,03366 + 0,02828 = 0,062]$$

Parcial 2 2024

Ej 2) Función de Tensión de Airy: $\phi(x,y) = X^2 y$

a) Fuerzas de cuerpo despreciables ($b_i = 0$) ¿El cuerpo se encuentra en equilibrio? Justifique.
Tensiones σ_{ij} se obtienen como: $\sigma_{xx} = \frac{\partial^2 \phi}{\partial y^2}$ $\sigma_{yy} = \frac{\partial^2 \phi}{\partial x^2}$ $\sigma_{xy} = -\frac{\partial^2 \phi}{\partial x \partial y} - b_x y - b_y x$

$$[\text{Equilibrio: } \frac{\partial \sigma_{ij}}{\partial x_j} + b_i = 0]$$

$$\sigma_{xx} = \frac{\partial^2 \phi}{\partial y^2} = 0 \quad \sigma_{yy} = \frac{\partial^2 \phi}{\partial x^2} = 2y$$

$$\left\{ \begin{array}{l} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + b_x = 0 \quad (1) \\ \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + b_y = 0 \quad (2) \end{array} \right.$$

$$\sigma_{xy} = -\frac{\partial^2 \phi}{\partial x \partial y} = -2x$$

$$(1) \quad 0 + 0 = 0 \quad \checkmark$$

$$(2) \quad -2 + 2 = 0 \quad \checkmark$$

Equilibrio

b) material elástico lineal homogéneo ($\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}$) y pequeñas deformaciones ϵ_{ij}
verifique que se cumple Compatibilidad por: $\frac{\partial^2 \epsilon_{xx}}{\partial y^2} + \frac{\partial^2 \epsilon_{yy}}{\partial x^2} = 2 \frac{\partial^2 \epsilon_{xy}}{\partial x \partial y}$

y λ y μ constantes:

$$[\text{Compatibilidad: } \frac{\partial^2 \epsilon_{xx}}{\partial y^2} + \frac{\partial^2 \epsilon_{yy}}{\partial x^2} = 2 \frac{\partial^2 \epsilon_{xy}}{\partial x \partial y}]$$

$$\text{Partiendo de } \sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij} \quad \text{Tomamos } i=j=m$$

$$\sigma_{mm} = \lambda \epsilon_{kk} \delta_{mm} + 2\mu \epsilon_{mm} \quad \text{Tomamos } \sigma_{mm} = \sigma_{kk} \quad \epsilon_{mm} = \epsilon_{kk}$$

$$\sigma_{kk} = \lambda \epsilon_{kk} 3 + 2\mu \epsilon_{kk}$$

$$\sigma_{kk} = \epsilon_{kk} (3\lambda + 2\mu) \rightarrow \left[\epsilon_{kk} = \frac{\sigma_{kk}}{3\lambda + 2\mu} \right] \quad (2)$$

② em ①

$$\sigma_{ij} = \lambda \left(\frac{\sigma_{kk}}{3\lambda + 2\mu} \right) \delta_{ij} + 2\mu \varepsilon_{ij} \rightarrow 2\mu \varepsilon_{ij} = \sigma_{ij} - \lambda \left(\frac{\sigma_{kk}}{3\lambda + 2\mu} \right) \delta_{ij}$$

$$\varepsilon_{ij} = \frac{1}{2\mu} \cdot \left(\sigma_{ij} - \frac{\lambda}{3\lambda + 2\mu} \sigma_{kk} \delta_{ij} \right)$$

Por inciso a) $\sigma_{kk} = 2y$
 $\sigma_{xx} = 0$
 $\sigma_{yy} = 2y$

$$\varepsilon_{xx} = \frac{1}{2\mu} \cdot \left(\sigma_{xx} - \frac{\lambda}{3\lambda + 2\mu} \cdot 2y \cdot 1 \right) \rightarrow \frac{-\lambda}{(3\lambda + 2\mu) \cdot \mu}$$

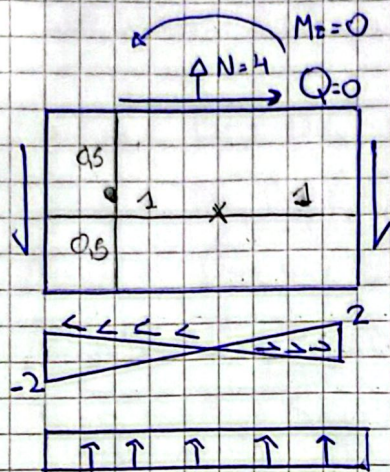
$$\varepsilon_{yy} = \frac{1}{2\mu} \cdot \left(\sigma_{yy} - \frac{\lambda}{3\lambda + 2\mu} \cdot 2y \cdot 1 \right) \rightarrow \frac{1}{2\mu} \left(2y - \frac{\lambda}{3\lambda + 2\mu} 2y \right) \rightarrow \left(\frac{1 - \lambda}{3\lambda + 2\mu} \right) \frac{1}{\mu} y$$

$$\varepsilon_{xy} = \frac{1}{2\mu} \cdot (-2x) = -\frac{1}{\mu} x$$

$$\left[\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} = \frac{2 \partial^2 \varepsilon_{xy}}{\partial x \partial y} \right] \rightarrow [0 + 0 = 0]$$

c) Verifique los valores de borde de fuerzas en el lado superior

Esfuerzos resultantes: normal N , esfuerzos de corte Q y momento M_z en el eje (Torques)



espesor = 2

de inercia I

$$\sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{bmatrix} = \begin{bmatrix} 0 & -2x \\ -2x & 2y \end{bmatrix}$$

y varía de -0.5 a 0.5 x varía de -1 a 1

Usamos Cauchy $T = \sigma \cdot m$

Carra superior $m = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\begin{bmatrix} T_x \\ T_y \end{bmatrix} = \begin{bmatrix} 0 & -2x \\ -2x & 2y \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\begin{matrix} T_x = -2x \\ T_y = 2y \end{matrix}$

Analizamos $y = 0.5$ y x quedará $T_{sup} = \begin{bmatrix} -2x \\ 1 \end{bmatrix}$

Carra inferior $m = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ $\begin{bmatrix} T_x \\ T_y \end{bmatrix} = \begin{bmatrix} 0 & -2x \\ -2x & 2y \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -1 \end{bmatrix} \Rightarrow \begin{matrix} T_x = 2x \\ T_y = -2y \end{matrix}$

Analizamos $y = -0.5$ y x quedará $T_{inf} = \begin{bmatrix} 2x \\ -1 \end{bmatrix}$

$N_{superior} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $N = \int_{x=-1}^{x=1} \int_{z=0}^{z=espesor} 1 dz dx \rightarrow \int_{-1}^1 z \Big|_0^{espesor} dx \rightarrow \int_{-1}^1 e dx \rightarrow$

$N = e \cdot x \Big|_{-1}^1 \rightarrow e \cdot (1 + 1) = 2 \cdot e = 4$

$Q = \text{Componente } \sigma_{xy} = \sigma_{yx} \text{ de Cauchy} \rightarrow e \cdot \int_{x=-1}^{x=1} \sigma_{xy} dx \rightarrow e \cdot \int_{-1}^1 -2x dx \rightarrow e \cdot \frac{-2x^2}{2} \Big|_{-1}^1$

$Q = e \cdot (1 - 1) = 0$

Moment = Brazo de palanca $\times$ Fuerza, M_z

$$M_z = e \int_{x=-1}^{x=1} (x T_y - y T_x) dx$$

$$T_y = \text{Presión normal} = 1 \quad T_x = \text{Presión tangencial} = 2$$

$$y = 0,5 \quad e = 2$$

$$M_z = 2 \int_{x=-1}^{x=1} (x \cdot 1 - 0,5 \cdot 2x) dx$$

$$M_z = 2 \int_{-1}^1 2x dx \rightarrow 2 \cdot \left. \frac{2x^2}{2} \right|_{-1}^1 \rightarrow 0$$

3) Verifique que o seguinte Tensor é isotrópico

$$A_{ijkl} = S_{ik} S_{jl}$$

Tensor Isotrópico: Aquel que não muda de forma quando se muda as coordenadas

$$A_{mnpq}' = B_{mi} B_{nj} B_{pk} B_{ql} A_{ijkl} \quad [A_{ijkl} = S_{ik} S_{jl}]$$

$$A_{mnpq}' = B_{mi} B_{nj} B_{pk} B_{ql} S_{ik} S_{jl}$$

$$A_{mnpq}' = B_{mk} B_{nl} B_{pk} B_{ql}$$

$$A_{mnpq}' = B_{mk} B_{pk} B_{nl} B_{ql}$$

$$A_{mnpq}' = S_{mp} S_{nq} \quad \rightarrow \quad A_{ijkl} = S_{ik} S_{jl}$$

$$4) \quad P \frac{D^2 u_i}{Dt^2} = \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j} + (\mu + \lambda) \frac{\partial \varepsilon_{jj}}{\partial x_i} + b_i$$

Cantidad de movimiento $P_i = \int_V P v_i dv$

Resultado de las presiones $f_i = \int_S T_i ds + \int_V b_i dv$

Formulas de Cauchy: $T_i = m_j \Pi_{ji}$

Balance de cantidad de movimiento: $\frac{D}{Dt} (P_i) = f_i$

Ecuación de continuidad $\frac{D\rho}{Dt} + \rho \frac{\partial v_i}{\partial x_i} = 0$

Velocidad $v_i = \frac{Du_i}{Dt}$

Tensor de deformación: $\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$

Deformación Volumétrica $\varepsilon_{kk} = \frac{\partial u_k}{\partial x_k}$

1: Partimos de ecuación de balance de cantidad de movimiento lineal

$$P \frac{Dv_i}{Dt} = \frac{\partial \sigma_{ij}}{\partial x_j} + b_i$$

notamos que v_i es la derivada material del desplazamiento u_i
 $v_i = \frac{Du_i}{Dt}$ notamos de nuevo que la derivada

$$P \frac{D^2 u_i}{Dt^2} = \frac{\partial \sigma_{ij}}{\partial x_j} + b_i$$

$$\frac{Dv_i}{Dt} = \frac{D^2 u_i}{Dt^2}$$

2: usamos la ley de Hooke

ley de Hooke: $\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}$

$$\frac{\partial \sigma_{ij}}{\partial x_j} = \lambda \frac{\partial \varepsilon_{kk}}{\partial x_j} \delta_{ij} + 2\mu \frac{\partial \varepsilon_{ij}}{\partial x_j}$$

$$\frac{\partial \sigma_{ij}}{\partial x_j} = \lambda \frac{\partial \varepsilon_{kk}}{\partial x_i} + 2\mu \frac{\partial \varepsilon_{ij}}{\partial x_j}$$

Tomamos $k=j$

$$\frac{\partial \sigma_{ij}}{\partial x_j} = \lambda \frac{\partial \varepsilon_{jj}}{\partial x_i} + 2\mu \frac{\partial \varepsilon_{ij}}{\partial x_j}$$

Paso 3: Transformar deformaciones en desplazamientos

Unas pequeñas deformaciones $\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$

Tomamos el término $2u \frac{\partial \epsilon_{ij}}{\partial x_j}$ y nos fijamos

$$2u \frac{\partial}{\partial x_j} \left(\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right)$$

$$u \left(\frac{\partial^2 u_i}{\partial x_j \partial x_j} + \frac{\partial^2 u_j}{\partial x_j \partial x_i} \right)$$

Conmutamos orden y aporamos 2º Term

$$u \left(\frac{\partial^2 u_i}{\partial x_j \partial x_j} + \frac{\partial}{\partial x_i} \left(\frac{\partial u_j}{\partial x_j} \right) \right)$$

$\frac{\partial u_j}{\partial x_j} = \text{div. volumen de } \epsilon_{jj}$

$$u \frac{\partial^2 u_i}{\partial x_j \partial x_j} + u \frac{\partial \epsilon_{jj}}{\partial x_i}$$

Paso 4: Reemplazamos todo

$$\frac{\partial \theta_{ij}}{\partial x_j} = \lambda \epsilon_{jj} + u \frac{\partial^2 u_i}{\partial x_j \partial x_j} + u \frac{\partial \epsilon_{jj}}{\partial x_i}$$

$$\frac{\partial \theta_{ij}}{\partial x_j} = u \frac{\partial^2 u_i}{\partial x_j \partial x_j} + (\lambda + u) \frac{\partial \epsilon_{jj}}{\partial x_i}$$

Paso 5: Ec. Original

$$\rho \frac{Dv_i}{Dt} = \frac{\partial \theta_{ij}}{\partial x_j} + b_i$$

$$\rho \frac{D^2 u_i}{Dt^2} = u \frac{\partial^2 u_i}{\partial x_j \partial x_j} + (\lambda + u) \frac{\partial \epsilon_{jj}}{\partial x_i} + b_i$$

$$5) \quad q = \begin{bmatrix} X_1 t^2 \\ X_2 \cos(t) \\ X_3 e^t \end{bmatrix} \quad v = \begin{bmatrix} X_1/t \\ X_2 \sin(t) \\ X_3 \end{bmatrix} \rightarrow \begin{matrix} X_1/t \\ X_2 \sin(t) \\ X_3 \end{matrix}$$

a) Derivada Temporal material del campo $q_i (Dq_i/Dt)$

$$\frac{Dq_i}{Dt} = \frac{\partial q_i}{\partial t} + \frac{\partial q_i}{\partial x_j} v_j \quad \frac{\partial q_i}{\partial t} = \begin{bmatrix} 2X_1 t \\ -X_2 \sin(t) \\ X_3 e^t \end{bmatrix}$$

$$\frac{\partial q_i}{\partial x_j} = \begin{bmatrix} \frac{\partial q_1}{\partial x_1} & \frac{\partial q_1}{\partial x_2} & \frac{\partial q_1}{\partial x_3} \\ \frac{\partial q_2}{\partial x_1} & \frac{\partial q_2}{\partial x_2} & \frac{\partial q_2}{\partial x_3} \\ \frac{\partial q_3}{\partial x_1} & \frac{\partial q_3}{\partial x_2} & \frac{\partial q_3}{\partial x_3} \end{bmatrix} = \begin{bmatrix} t^2 & 0 & 0 \\ 0 & \cos(t) & 0 \\ 0 & 0 & e^t \end{bmatrix}$$

$$\frac{Dq_i}{Dt} = \begin{bmatrix} 2X_1 t \\ -X_2 \sin(t) \\ X_3 e^t \end{bmatrix} + \begin{bmatrix} t^2 & 0 & 0 \\ 0 & \cos(t) & 0 \\ 0 & 0 & e^t \end{bmatrix} \cdot \begin{bmatrix} X_1/t \\ X_2 \sin(t)/\cos(t) \\ X_3 \end{bmatrix}$$

$$\frac{Dq_i}{Dt} = \begin{bmatrix} 2X_1 t \\ -X_2 \sin(t) \\ X_3 e^t \end{bmatrix} + \begin{bmatrix} X_1 t \\ X_2 \sin(t) \\ e^t X_3 \end{bmatrix} = \begin{bmatrix} 3X_1 t \\ 0 \\ 2X_3 e^t \end{bmatrix}$$

b) El flujo $\frac{Dq_i}{Dt}$ a través de la superficie de un volumen fijo en los ejes cartesianos y de volumen A

$$\iint_S \frac{Dq_i}{Dt} n_i dS$$

$$\iint_S \frac{Dq_i}{Dt} n_i dS = \iiint_V \frac{\partial}{\partial x_i} \left(\frac{Dq_i}{Dt} \right) dV \quad \frac{\partial}{\partial x_i} \left(\frac{Dq_i}{Dt} \right) = \begin{pmatrix} 3t \\ 0 \\ 2e^t \end{pmatrix}$$

$$\iiint_V (3t + 0 + 2e^t) dV$$

$$(3t + 2e^t) \iiint_V dV \rightarrow (3t + 2e^t) \cdot V \quad V_{\text{volumen}} = A$$

$$[(3t + 2e^t) \cdot A]$$